

Beyond the Specialist: Analyzing Cross-Domain Zero-Shot Embeddings of Time Series Foundation Models*

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December 29, 2025

Abstract

The emergence of Time Series Foundation Models (TSFMs) promises a shift toward universal, zero-shot forecasting capabilities. However, their effectiveness across diverse domains remains an open question. In this study, we conduct a geometric audit of six TSFMs (TimesFM, Moirai, Wav2Vec 2.0, ECGFounder, Astromer, FinCast) across four distinct domains: Finance (S&P 500 stocks), Healthcare (PTB-XL), Audio (ESC-50), and Astronomy (StarEmbed). By deploying a multi-head classification suite, ranging from logistic regression to multi-layer perceptrons, we benchmark these large-scale models against random embeddings and lightweight, domain-specific features. We find that TSFMs demonstrate strong positive transfer in deterministic, physical domains, yet fail to significantly outperform random noise in high-entropy financial environments. Furthermore, we observe that domain-specific foundation models do not consistently outperform generalist architectures. Notably, lightweight custom feature extractors frequently rival or outperform massive pre-trained models, achieving 90.3% accuracy in astronomy tasks where TSFMs capped at 67.9%, challenging the assumption that scale is the only path to performance. We conclude that current pre-training objectives successfully encode morphological and periodic laws but struggle to disentangle stochastic signal from noise, necessitating a re-evaluation of zero-shot transfer in non-stationary domains.

1 Introduction

Time Series Foundation Models (TSFMs) are an increasingly popular class of models that promise to deliver predictive and classification power with little to no fine-tuning necessary. Recent advances have suggested that pre-trained forecasting models can serve as useful zero-shot feature extractors (Auer et al. 2025) for classification. However, the universality of these models in forecasting tasks has been challenged by Olivares et al. (2025), which emphasizes the difficulty of cross-domain learning and notes that the accuracy of statistical baselines is frequently underreported. We extend this line of inquiry, which has largely focused on generalist architectures, by including domain-specific models in our benchmark. This inclusion of specialists allows us to quantify the trade-off between massive scale and domain-focused training.

We investigate the zero-shot, cross-domain capabilities of these models in four different domains: finance (S&P 500 stocks), healthcare (electrocardiography), audio (environmental sounds), and astronomy (variable star light curves). From each dataset, we extract embeddings and apply a suite of classification heads, including logistic regression, nearest centroid, k-nearest neighbours, and a multi-layer perceptron. This multi-probe approach allows us to characterize the geometric structure of the embedding spaces, thereby diagnosing how different pre-training objectives shape the representation topology across diverse signal types.

A key test of foundation model quality is whether pre-training on domain data produces representations that capture domain-relevant features. For instance, FinCast, trained on financial data, should implicitly encode momentum indicators, volatility patterns, and mean-reversion dynamics that our custom RSI/MACD features explicitly compute. If specialists fail to match custom features, this suggests their pre-training objectives are misaligned with downstream utility.

*Code and data are available at: <https://github.com/ploewen/CDTSFM>.

2 Data

We consider 4 datasets spanning the audio, astronomical, finance, and medical domains. Specifically, we utilize the **PTB-XL** (Wagner et al. 2020) dataset for electrocardiography (ECG), **ESC-50** (Piczak 2015) for environmental audio classification, daily stock returns from **S&P 500 constituents**, and **StarEmbed** (W. Li et al. 2025) for astronomical light curve classification. This multi-domain approach allows us to test TSFM embeddings against various signal types, with each dataset posing a distinct challenge.

We provide data examples in Figure 1 and summarize the descriptive statistics and downstream tasks for each dataset in Table 1. Comprehensive descriptions of the datasets, including preprocessing steps and label distributions, are provided in Appendix A.

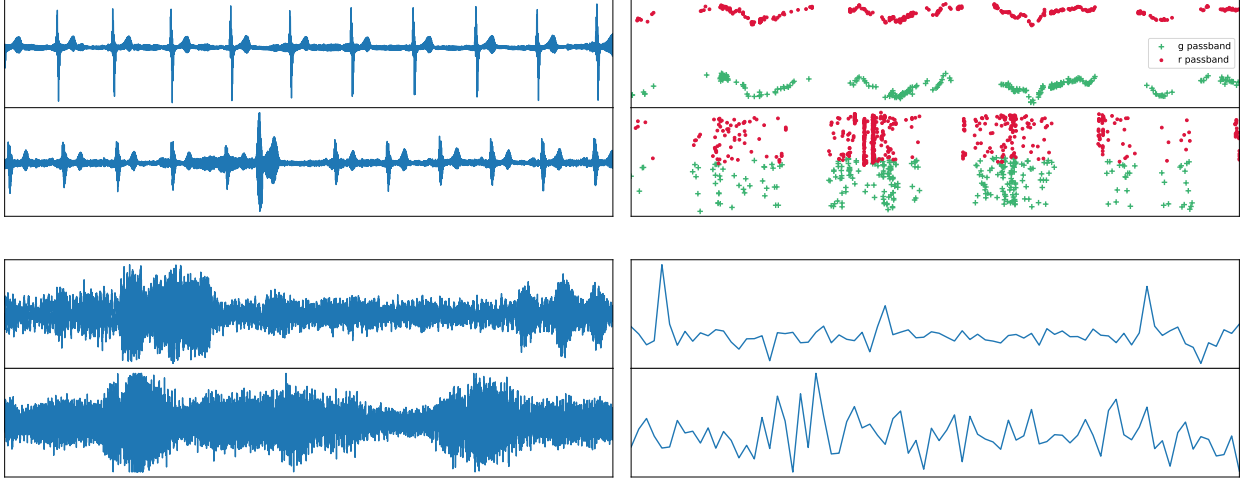


Figure 1: Examples from PTB-XL (top left), StarEmbed (top right), ESC-50 (bottom left), and S&P 500 constituents (bottom right).

Table 1: Dataset summaries

Dataset	Challenge	Number of classes	Series length	Sample size
PTB-XL	Highly periodic and high inter-patient variability	6	80000	21799
StarEmbed	Multiband and irregularly sampled with variable series length	7	Variable	41498
ESC-50	High frequency with similar classes (e.g. rain vs crackling fire)	50	12000	2000
S&P 500	Low signal to noise ratio, highly stochastic	2	80	359312

3 Models

We evaluate six diverse models, categorized into two different groups: generalist models designed for universal forecasting and classification, and specialist models trained in specific data domains. Our generalist models are **Moirai** (Woo et al. 2024) and **TimesFM** (Das et al. 2024), which are compared to the domain specialists **Astromer** (Donoso-Oliva et al. 2022), **ECGFounder** (J. Li et al. 2025), **FinCast** (Zhu et al. 2025), and **Wav2Vec 2.0** (Baeviski et al. 2020). Models were selected to prioritize open-weight availability and to span the primary architecture paradigms in time series learning tasks.

3.1 General Models

Moirai is a masked-encoder based time series forecasting model developed by Salesforce Research to perform across domains (Woo et al. 2024). Moirai utilizes multiple patch size projection layers and an Any-variate Attention mechanism, enabling it to treat data with varying temporal patterns and dimensionality seamlessly. Furthermore, the model addresses the issue of probabilistic forecasting by employing a mixture of parametric distributions.

TimesFM is a model developed by Google Research (Das et al. 2024), employing a Decoder-Only paradigm. TimesFM draws inspiration from Large Language Models (LLMs) and processes non-overlapping patches of the series into tokens with a Multi-Layer Perceptron, using a Transformer architecture to autoregressively predict following patches. The pre-training set consists of a mixture of 100 billion time points, composed of both real-world data (Google Trends, Wikipedia Pageviews) and synthetic data, generated through statistical models and physical simulations.

3.2 Domain Specific Models

While generalist models aim for universality, **Astromer** (Donoso-Oliva et al. 2022) demonstrates the power of domain-focused pre-training. It is a self-supervised, Transformer-based encoder-decoder designed specifically for astronomical light curve data, representing the brightness of celestial objects. ASTROMER has been developed to handle the irregular sampling inherent to telescopic observation (e.g., gaps due to daylight or weather).

ECGFounder (J. Li et al. 2025) is a large-scale encoder designed for electrocardiogram (ECG) analysis. Trained on over 10 million 12-lead ECG recordings, it learns to identify 150 distinct cardiac abnormalities. ECGFounder employs a RegNet (ResNet-based) backbone optimized for capturing the precise morphological features of cardiac cycles (such as QRS complex width and ST-segment elevation).

Wav2Vec 2.0 (W2V) (Baevski et al. 2020) is a model developed by Meta AI, employing a dual-stage architecture. It first employs a multi-layer Convolutional Neural Network (CNN) to process raw waveforms and extract latent feature representations, followed by a context network which uses a standard Transformer architecture. Wav2Vec 2.0’s key contribution is a Contrastive Learning objective combined with Product Quantization. This architecture builds latent representations by concatenating entries of a codebook, allowing it to capture complex frequency-domain dependencies and local temporal structures inherent to sound.

FinCast (Zhu et al. 2025) is a foundation model tailored for financial time series forecasting. It has been trained exclusively on large-scale financial data, encompassing diverse asset classes (equities, forex, commodities) and frequencies. The model uses input tokenization, followed by Causal Attention, sparse Mixture of Experts (MOE) blocks, and a Decoder. FinCast aims to capture domain-specific latent dynamics like volatility clustering and regime changes that generic models may fail to prioritize.

4 Methodology

To evaluate the intrinsic geometry of TSFM representations, we adopt a fixed-feature extraction protocol. For each input series $\mathbf{y}$, we compute a latent embedding $\mathbf{z} = f_\phi(\mathbf{y})$ using pre-trained model parameters ϕ . Crucially, ϕ remains frozen throughout our experiments to prevent task-specific adaptation. We then train a set of lightweight classification heads $g_\psi(\mathbf{z})$ to map these embeddings to target labels. This decoupling allows us to isolate the quality of the zero-shot embeddings from the capacity to learn new features. In the case of the StarEmbed dataset, which contains two variates, the embeddings generated for each band are concatenated following Auer et al. (2025).

Our classification heads consist of **logistic regression** to test for linearly separable manifolds, **nearest centroid** for spherical clusters, **k-nearest neighbours** (with $k=5$ and distance weighted classification) for non-linear local neighbourhoods, and a **multi-layer perceptron** (with one hidden layer of dimension 1024) for complex non-linear relationships (see Appendix B for formal definitions).

To assess model performance, we report four standard metrics: Accuracy, Precision, Recall, and F_1 score. While Accuracy serves as a primary metric for comparisons, the other metrics allow us to account for imbalanced classes. In the case of multinomial prediction, we take Precision, Recall and, F_1 score to be the weighted average of the same metrics computed in a binary setting for each class.

Finally, for each data-model pair we estimate the intrinsic dimension using 2-nearest neighbours (Facco et al. 2017), and anisotropy by computing the average cosine similarity, to analyze the embedding geometry.

5 Experiments and Results

5.1 Experiment Setup

All experiments were conducted within a standardized benchmarking pipeline to ensure fair comparison.

Preprocessing. To reduce any information leakage, training and testing data were immediately split. Dimensionality reduction (PCA) and standardization were fitted only on the training split and applied to the testing split. Where appropriate, downsampling was applied, greatly reducing the size of the data files and increasing computational speed while retaining the same information.

Baselines. We compare all pre-trained models against two basic controls. A custom embedding has been generated for each dataset, computing various statistical features in both the time and frequency domains (see Appendix C for the list of features included). These embeddings serve as a baseline for the other models, demonstrating the explainability of light-weight statistical methods compared to the much more computationally expensive and pre-trained models. We also include random embeddings, which are constructed by generating vectors of length 512 from independent Uniform(0,1) random variables. This random embedding allows us to measure class balance bias.

5.2 Results

Tables 2 and 3 summarize the results of our experiment. Complete tables of results are given in Appendix D. Table 2 demonstrates a high variance in transferability between signal types. Notably, the lightweight custom features achieved the highest accuracy in 2 out of 4 datasets.

In the astronomy domain, the custom baseline reached 90.3% accuracy, far outperforming the nearest model (Astromer, 67.9%). Surprisingly, ECGFounder achieved the best accuracy at 47.5% in the audio classification task, outpacing the specialist Wav2Vec (28.2%). Performance in the S&P 500 dataset remained uniformly low. The best performing foundation models (Wav2Vec, Astromer, 52.4%) achieved a slight edge over the random embeddings (52.0%). Within the ECG classification task, the top models were Moirai (65.7%), ECGFounder (63.5%), and TimesFM (62.2%), exhibiting positive transferability in the generalist models.

Model	S&P 500	PTB-XL	ESC-50	StarEmbed
<i>Generalist Models</i>				
TimesFM	0.513	0.622	0.370	0.676
Moirai	0.518	0.657	0.328	0.676
<i>Domain Specialists</i>				
Wav2Vec	<u>0.524</u>	0.470	0.282	0.676
Astromer	0.524	0.557	0.158	<u>0.679</u>
FinCast	0.523	0.437	0.023	0.676
ECGFounder	0.516	<u>0.635</u>	0.475	0.676
<i>Baselines</i>				
Custom	0.542	0.557	<u>0.455</u>	0.903
Random	0.520	0.437	0.033	0.676

Table 2: Summary of best accuracy achieved by each model across all datasets (using best-performing classification head). Best results are **bold**, second best are underlined.

Table 3 characterizes the geometric structure of embeddings by showing which classification heads achieved optimal performance for each model across all datasets.

The logistic regression head achieved top performance in 21 of 32 model-dataset combinations, with the multi-layer perceptron achieving second place with 10 top results. Importantly, k-nearest neighbours only achieved one top result, with the nearest centroid never achieving top performance. The custom features distinguish themselves from the time series models as they uniquely relied on the logistic regression head to achieve top results, indicating a strongly linearly separable embedding space.

Model	LogReg	k-NN	MLP	Centroid
<i>Generalist Models</i>				
TimesFM	2	1	1	0
Moirai	3	0	1	0
<i>Domain Specialists</i>				
Wav2Vec	3	0	1	0
Astromer	2	0	2	0
FinCast	3	0	1	0
ECGFounder	2	0	2	0
<i>Baselines</i>				
Custom	2	0	2	0
Random	4	0	0	0
Total	21	1	10	0

Table 3: Distribution of best-performing classification heads for each model across 4 datasets. Bold indicates the dominant head(s) for each model (ties shown as multiple bold values).

Table 4 presents a partial geometric analysis of the embedding spaces (the full analysis for all data-model combinations can be found in Appendix D.5). We computed the intrinsic dimension and anisotropy (average cosine similarity) to evaluate the topological structure of the learned representations.

The random baseline for all datasets exhibits high intrinsic dimension (122.5-159.9) and moderate anisotropy ($\approx 75\%$). While ECGFounder maintained a moderately high intrinsic dimension, explaining its success in heart rhythm classification, other specialists underwent severe manifold collapse. Most notably in the financial domain, despite the high-entropy baseline (with ID 159.9), all other models were severely compressed (collapse ratio 91.2%-97.5%).

6 Discussion

6.1 Model Performance

Our results reveal a distinct dichotomy in the efficacy of TSFMs. For heart rhythm classification, Moirai and TimesFM demonstrated strong positive transfer achieving results competitive with the domain specialist ECGFounder. However, within the light curve and stock classification tasks, they performed on par with the random noise embedding, demonstrating negative transfer when compared to the specialist models. Furthermore, the dominance of the Custom features in all but the Heart Rhythm domain challenges the “universal” applicability of pre-trained models.

The results in Table 1 show that TimesFM and Moirai failed to outperform the random embeddings, showing a lack of generalizability and suggesting that features learned during pretraining on a period or clean data do not map to the highly volatile, chaotic dynamics of financial markets.

In contrast, Table 2 demonstrates the clear utility of foundation models in domains determined by highly periodic data. Moirai, TimesFM, and ECGFounder outperformed all other models by a significant margin. This result suggests that large-scale pretraining captures universal morphological features such as waveform periodicity and shape.

Domain	Model	Embed Size	Intrinsic Dim	Anisotropy
Finance (S&P 500)	Random Baseline	512	159.9	0.750
	Custom Baseline	18	5.3	0.951
	FinCast (Specialist)	1280	4.0	0.999
	Moirai (Generalist)	384	11.2	0.748
Astronomy (Star)	Random Baseline	512	146.6	0.751
	Custom Baseline	54	5.6	0.775
	Astromer (Specialist)	513	4.9	0.996
	Moirai (Generalist)	768	10.0	0.838
Medical (PTB-XL)	Random Baseline	512	136.9	0.750
	Custom Baseline	18	3.4	0.809
	ECGFounder (Specialist)	1024	19.6	0.757
	Moirai (Generalist)	384	13.7	0.970
Audio (ESC-50)	Random Baseline	512	122.5	0.750
	Custom Baseline	36	3.5	0.743
	Wav2Vec 2.0 (Specialist)	769	2.5	0.998
	Moirai (Generalist)	385	2.0	1.000

Table 4: Comparison of embedding space topology across domains. Note the contrast between the rich dimensionality of medical embeddings ($ID \approx 19.6$) versus the collapsed financial embeddings ($ID \approx 4.0$), despite large embedding sizes.

Surprisingly, ECGFounder, trained on heartbeat data, achieved the highest accuracy (47.5%) on the audio classification task, outperforming the generalists TimesFM (37.0%) and Moirai (32.8%) as well as the specialist Wav2Vec (25.7%). This points to a successful transfer between domains. This may be due to the similarities in tasks, where both audio and heartbeats can be understood due to their frequency domain properties.

In the astronomy domain, the Custom features drastically outperformed all other models, achieving 90.3% accuracy with the MLP. In comparison, the second-best model was Astromer reaching only 67.9% accuracy. This 22% performance gap indicates that current embeddings fail to capture the high-precision phase information required to correctly distinguish star classes, whereas a statistical model successfully captures the correct information.

6.2 Embedding Geometry

A universal finding across all experiments, and especially in the astronomy domain, is that out of all classification heads, the centroid classifier demonstrated by far the least competitive results. This is consistent with the extreme anisotropy observed in our analysis in Table 13, where model anisotropy is above 60%. As noted by Ethayarajh (2019), this suggests that TSFMs do not represent data as compact, spherical sets but rather collapse the data into a narrow cone.

Within the highly stochastic financial domain, we see a wide range of geometric metrics. The financial specialist FinCast embeds the data in a low-dimensional, highly anisotropic space, while ECGFounder utilizes a high-dimensional space with extremely low anisotropy. This demonstrates the difficulty of modelling stochastic patterns.

We can see that ECGFounder and Moirai consistently demonstrate lower anisotropy than other models, with higher intrinsic dimension. This suggests that these models utilize a more complex and non-linear geometry to construct embeddings, allowing the models to model a wider range of signals, leading to their positive transfer capabilities. Conversely, Astromer demonstrates lower intrinsic dimension and high anisotropy consistent with its low transferability. The custom features demonstrate that a low intrinsic dimension may not be fatal; in fact, when paired with low anisotropy, the custom features compete with all top models.

We propose that low anisotropy is a necessary condition for effective and adaptive representation learning, regardless of the dimensionality of the embedding.

7 Conclusion

In this study, we have conducted a rigorous geometric audit of state-of-the-art Time Series Foundation Models (TSFMs) to evaluate their performance and their promise of zero-shot transferability. By evaluating the performance of different classifiers on the embeddings of models, general models Moirai, TimesFM, as well as specialized models Astromer, FinCast, ECGFounder, and Wav2Vec 2.0, against random and stochastic-based models, we arrive at three main conclusions.

We find that the “universality” of current foundation models is conditional. TSFMs exhibit strong transferability in domains governed by deterministic laws (e.g., within the PTB-XL and ESC-50 datasets), where the signal is generally clean and periodic. However, in stochastic, high-entropy domains such as finance, these large models fail to generalize, performing similarly to random noise embeddings. This suggests that TSFMs tend to learn safe, low-dimensional embeddings rather than disentangling signal from noise.

Secondly, our results challenge the assumption that simply pretraining on domain-specific data guarantees top performance. The finance specialist’s FinCast embeddings were dominated by noise, as evidenced by their proximity to the random baseline. In contrast, ECGFounder generalized extremely well to the audio classification task. This suggests that the structure of the data may be a more important factor than the semantic label of the domain.

Geometrically, we find that TSFM transferability is conditional on its geometric properties. We observe two distinct regimes for successful representation. Firstly, rich manifolds with high intrinsic dimension and low anisotropy, as exemplified by ECGFounder, manage to capture complex relationships in the data. The drawback of this regime is shown in its failure within the finance domain, where it likely utilized its adaptability to model noise. Secondly, we have low dimension and low anisotropy, as exemplified by the custom features most notably in Astronomy, which separates the data by extracting important factors. Crucially, failure is universally characterized by high anisotropy. This embeds the data into narrow cones, failing to capture the complexity of the data.

Finally, our study challenges the assumption that scale guarantees performance. In specialized tasks such as variable star classification, a custom model using stochastic features drastically outperformed the best foundation model. Despite the paradigm shift towards large-scale foundation models, light-weight, task-specific feature extractors remain highly competitive in high-entropy domains and should not be discarded in favor of expensive generalist architectures. Consequently, we argue that the proper direction for improvement in transferability lies not in increasing data volume or pre-training domain, but in developing pre-training objectives that balance dimensionality reduction and topological adaptability.

8 Limitations

While our study provides evidence for domain-specific performance gaps in TSFMs, several methodological constraints should be considered when interpreting our findings.

Our custom features encode decades of domain expertise (e.g., RSI and MACD for finance, Lomb-Scargle periodograms for astronomy). However, these are not “privileged” baselines but rather represent the knowledge that domain specialists should have acquired during pre-training.

Furthermore, our study lacks systematic ablation experiments that would isolate the contribution of specific design choices. Key ablations not performed include embedding extraction strategies (mean vs max pooling, layer selection), input sequence length sensitivity, custom feature subset analysis, PCA dimensionality impact, classification head capacity scaling, or training regime variations. These missing ablations mean our results represent lower bounds on TSFM performance rather than performance ceilings, and that the true capabilities of each model might exceed our measurements.

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A Dataset Descriptions

PTB-XL is a large publicly available electrocardiography dataset (Wagner et al. 2020), containing 21799 ECG 10 second readings using 12 leads from 18869 patients. The data has been expert annotated, conforming to the SCP-ECG standard. Included annotations are “NORM” (Normal ECG), “MI” (Myocardial Infarction), “STTC” (ST/T Change), “CD” (Conduction Disturbance), and “HYP” (Hypertrophy). Additionally, an “OTHER” annotation was added for observations that were not given an annotation. The ECG recordings were recorded at a frequency of 500 Hz and downsampled to 100Hz for processing purposes.

	Train Count	Train (pct)	Test Count	Test (pct)
CD	1957	10.0	222	10.1
HYP	2387	12.2	262	11.9
MI	4180	21.3	471	21.4
NORM	8548	43.6	961	43.7
OTHER	371	1.9	40	1.8
STTC	2158	11.0	242	11.0
Total	19601	100.0	2198	100.0

Table 5: Label distribution for PTB-XL.

ESC-50 (Piczak 2015) is a dataset containing 2000 environmental audio recordings. The recordings are uniformly sampled at 44.1kHz with a fixed length of 5 seconds. Each recording is organized into one of 50 semantic classes, with 40 examples per class. The available classes can be categorized as animals, natural soundscapes, water sounds, human non-speech sounds, interior/domestic sounds, exterior/urban sounds (Piczak 2015). The data is prearranged by the author into 5 folds for cross validation. We reserve one of these splits to be our testing dataset, using the remaining 4 splits as testing data. This ensures balanced training and testing sets.

The **S&P 500** (Standard and Poor’s 500) is an index fund comprised of the 500 largest stocks. Our dataset is generated from the daily returns of each constituent stock of the index, from January 1st 2000 to November 1st, 2025. It has been sourced through the Yahoo Finance API (Yahoo Finance 2025) by usage of the yfinance Python package (Aroussi 2025). To pre-process the data we adapt the methodology put forward in (Krauss, Do, and Huck 2017). We do so by creating windows of 500 days, reserving the first 349 days of the window for our training set and the last 150 days for our testing set. We have split the data in this manner to ensure there is no data leakage, adopting purging (no overlap between training and testing sets) and embargo (removing any observation directly following a training set) methods from (De Prado 2018). Financial literature suggests that forecasting based on classifying the direction of the return generally performs better than level estimation (Leung, Daouk, and Chen 1999). To this end, we set the input variables of our models to be composed of the returns from sliding window of 80 days (t_0 to t_{79}) with the output variable set as 1 or 0 if the following day’s return (at t_{80}) is above or below the window’s median return respectively. The median was calculated strictly on the input window t_0 to t_{79} .

	Train Count	Train (pct)	Test Count	Test (pct)
1	22438	67.6	5610	67.6
13	367	1.1	92	1.1
2	3410	10.3	853	10.3
4	1659	5.0	415	5.0
5	3849	11.6	962	11.6
6	352	1.1	88	1.1
8	1122	3.4	281	3.4
Total	33197	100.0	8301	100.0

Table 6: Label distribution for StarEmbed

The **StarEmbed** dataset is a multi-band light curve dataset designed to benchmark astronomical and general TSFMs on astronomical time series data (W. Li et al. 2025). The dataset is composed of light curves selected from release 23 (DR23) of the ZTF astronomical survey (Bellm et al. 2018). Star class labels have been expertly labelled in the Catalina Surveys Periodic Variable Star Catalog (CSPVS) (Drake et al. 2014).

The dataset is split into training, validation and testing datasets. Time series for green and red light curves are given, along with time stamps formatted in Modified Julian Dates (MJD), and a string identifying the class of the star. Several other variables are included but will not be used for our analysis.

The stars were first pre-classified using multivariate kernel density estimation (Scott 1992), followed by a classification done by visual inspection to assess their final class (Drake et al. 2014). In total, there are 6 different star classes present in the dataset: “CVn”, “EW”, “EA”, “LPV”, “RRab”, “RRc”, “RRd”, and “RS”. All classes are present in approximately equal proportion in all three splits, however, some classes have many fewer observations than others, for example, LPV and EW classes which contain 360 and 27075 observations respectively (Drake et al. 2014).

	Train Count	Train (pct)	Test Count	Test (pct)
0	145459	51.0	38798	52.4
1	139877	49.0	35178	47.6
Total	285336	100.0	73976	100.0

Table 7: Label distribution for S&P 500 stocks

B Classification Heads

B.1 Logistic regression

Logistic regression defines $\psi = \{\beta, \mathbf{b}\}$ where the probability is modeled via the softmax function:

$$P(\ell = k \mid \mathbf{z}) = \frac{\exp(\beta_k^\top \mathbf{z} + b_k)}{\sum_{j \in \mathcal{L}} \exp(\beta_j^\top \mathbf{z} + b_j)}$$

B.2 Nearest Centroid

Nearest centroid sets, $\psi = \{\mu_k\}_{k \in \mathcal{L}}$, where μ_k is the mean vector for class k . The probability is derived from the negative Euclidean distance:

$$P(\ell = k \mid \mathbf{z}) = \frac{\exp(-\|\mathbf{z} - \mu_k\|_2^2)}{\sum_{j \in \mathcal{L}} \exp(-\|\mathbf{z} - \mu_j\|_2^2)}$$

B.3 k-Nearest Neighbours

Unlike linear classification methods which assume that there is a global hyperplane that separates classes, k-NN estimates probabilities based on local topology. Let $\mathcal{N}_k(\mathbf{z})$ denote the set of indices of the k training samples closest to the embedding $\mathbf{z}$ according to the Euclidean metric. We then set $w_i = \frac{1}{\|\mathbf{z} - \mathbf{z}_i\|_2}$ to be the weight of the i -th neighbor. The posterior probability for class c is estimated as the normalized weighted sum of neighbors belonging to that class:

$$P(\ell = c \mid \mathbf{z}) = \frac{\sum_{i \in \mathcal{N}_k(\mathbf{z})} w_i \cdot \mathbb{I}(\ell_i = c)}{\sum_{j \in \mathcal{N}_k(\mathbf{z})} w_j}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

B.4 Multi-Layer Perceptron

We define a MLP as a function parametrized by $\psi = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \sigma(\cdot)\}$ where $\mathbf{W}_1, \mathbf{W}_2$ are matrices, $\mathbf{b}_1, \mathbf{b}_2$ are vectors and $\sigma(\cdot)$ is a non-linear activation function. First the embedding $\mathbf{z} \in \mathbb{R}^d$ is projected into a hidden representation $\mathbf{h} \in \mathbb{R}^m$ (in our case $m = 1024$) by:

$$\mathbf{h} = \sigma(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1).$$

This hidden state is then projected into a logit vector $\mathbf{o} \in \mathbb{R}^K$ for the K classes by:

$$\mathbf{o} = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2.$$

From this logit vector the posterior probabilities are obtained from a softmax activation:

$$P(\ell = k \mid \mathbf{z}) = \frac{\exp([\mathbf{W}_2 \mathbf{h} + \mathbf{b}_2]_k)}{\sum_{j \in \mathcal{L}} \exp([\mathbf{W}_2 \mathbf{h} + \mathbf{b}_2]_j)}$$

C Extracted Features

Table 8: Detailed Description of Extracted Features

Category	Feature Name	Description
General Statistics (Applied to all models)		
Distribution	Mean, Median	Measures of central tendency.
	Std. Deviation	Measure of data dispersion/spread.
	Min, Max	The extreme values of the time series window.
	Peak-to-Peak (PTP)	The absolute range ($Max - Min$).
	Skewness	Measure of asymmetry in the data distribution.
	Kurtosis	Measure of the "tailedness" of the distribution.
Energy	RMS	Root Mean Square; represents the average power of the signal.
	AUC	Area Under the Curve; sum of absolute values.
Dynamics	Zero Crossing Rate (ZCR)	The rate at which the signal changes sign (smoothness/noise proxy).
	Mean Crossing Rate (MCR)	The rate at which the signal crosses its own mean.
	Autocorrelation (Lag 1)	Correlation of the signal with itself shifted by one time step (persistence).
	Dominant Frequency	The frequency component with the highest magnitude in the FFT spectrum.
Finance / Technical Analysis (Applied to S&P 500)		
Momentum	RSI (14)	Relative Strength Index; measures speed/change of price movements (Overbought/Oversold).
Trend	MACD Difference	Moving Average Convergence Divergence; indicates trend direction and strength.
Volatility	Bollinger Band Width	Difference between the Upper and Lower Bollinger Bands (2σ); measures volatility.
Growth	Simple Return	The percentage change from the first to the last point in the window.
Medical / HRV (Applied to PTB-XL)		
HRV (Time)	RMSSD	Root mean square of successive RR interval differences; estimates vagal tone.
	MeanNN	The mean of the RR intervals (inverse of heart rate).
	SDNN	Standard deviation of NN intervals; estimates overall HRV.
	pNN50	Percentage of successive RR intervals differing by $> 50\text{ms}$.
Audio / Spectral (Applied to ESC-50)		
Timbre	MFCCs (13)	Mel-frequency cepstral coefficients; represents the spectral envelope (timbre).
Brightness	Spectral Centroid	The center of mass of the spectrum; correlates with the perception of "brightness".
Texture	Spectral Contrast	Difference between peaks and valleys in the spectrum; captures harmonic content.
Continued on next page		

Table 8 – continued from previous page

Category	Feature Name	Description
Astronomy / Light Curves (Applied to StarEmbed)		
Morphology	Amplitude	Half the difference between the median of the top 5% and bottom 5% magnitudes.
	Anderson-Darling	Test statistic for normality; high values indicate non-Gaussian behavior.
	BeyondNStd ($1\sigma, 2\sigma, 3\sigma$)	Percentage of points lying beyond N standard deviations from the weighted mean.
	Cusum	Range of the cumulative sum of the light curve; detects persistent shifts.
	Eta	Ratio of mean successive difference to variance; distinguishes trends from noise.
	Inter-Percentile Range	Difference between the 75th and 25th percentiles (Robust spread).
	Linear Trend	Slope of the linear regression fitted to the light curve.
	Mag. Percent Ratio	Fraction of points inside a specific quantile range (measures outlier dominance).
	Max Slope	The maximum absolute slope between two consecutive observations.
	Stetson K	Robust kurtosis measure designed for irregularly sampled light curves.
Periodicity	Freq1 / Freq2	Top two dominant frequencies identified by the Lomb-Scargle periodogram.
	Stetson J	Robust correlation coefficient between two observations (or bands) close in time.
	Flux Percentile Ratios	Ratios of flux percentiles (e.g., F_{60}/F_{40}) defining the shape of the folded curve.

D Results

D.1 Financial Forecasting

The S&P-500 constituents data serves as a binary classification problem. Stock return data is highly stochastic with low signal to noise ratio. The models must be able to detect and communicate the signal while taking into account the highly unpredictable nature of the returns.

Classifier	Metric	Wav2Vec	Astromer	FinCast	ECGFounder	TimesFM	Moirai	Custom	Random
k-NN	Accuracy	0.500	0.502	0.486	0.503	0.513	0.498	<u>0.512</u>	0.500
	Precision	0.501	0.502	0.501	0.503	0.514	0.503	<u>0.513</u>	0.501
	Recall	0.500	0.502	0.486	0.503	0.513	0.498	<u>0.512</u>	0.500
	F1	0.500	0.502	0.440	0.503	0.514	0.497	<u>0.512</u>	0.500
Logistic	Accuracy	<u>0.524</u>	0.524	0.523	0.516	0.508	0.518	0.542	0.520
	Precision	<u>0.519</u>	0.501	0.502	0.511	0.503	0.492	0.540	0.498
	Recall	<u>0.524</u>	0.524	0.523	0.516	0.508	0.518	0.542	0.520
	F1	<u>0.510</u>	0.375	0.393	0.505	0.498	0.412	0.540	0.409
Centroid	Accuracy	<u>0.512</u>	0.505	0.492	0.507	0.496	0.483	0.532	0.497
	Precision	<u>0.514</u>	0.507	0.501	0.512	0.497	0.500	0.534	0.498
	Recall	<u>0.512</u>	0.505	0.492	0.507	0.496	0.483	0.532	0.497
	F1	<u>0.512</u>	0.506	0.480	0.505	0.496	0.426	0.532	0.497
MLP	Accuracy	0.513 (0.002)	0.514 (0.004)	<u>0.520 (0.003)</u>	0.506 (0.002)	0.511 (0.000)	0.497 (0.020)	0.535 (0.003)	0.501 (0.002)
	Precision	0.510 (0.001)	0.503 (0.002)	<u>0.503 (0.003)</u>	0.507 (0.002)	<u>0.511 (0.001)</u>	0.507 (0.005)	0.534 (0.001)	0.502 (0.001)
	Recall	0.513 (0.002)	0.514 (0.004)	<u>0.520 (0.003)</u>	0.506 (0.002)	<u>0.511 (0.000)</u>	0.497 (0.020)	0.535 (0.003)	0.501 (0.002)
	F1	0.509 (0.003)	0.476 (0.015)	<u>0.427 (0.024)</u>	0.506 (0.002)	<u>0.511 (0.001)</u>	0.408 (0.010)	0.531 (0.002)	0.502 (0.002)

Table 9: Classification results for S&P 500 dataset. Best results are **bold**, second best are underlined. The MLP are run with 3 seeds (0, 1, 2) and results are reported as Mean (Standard Deviation).

D.2 Heart Rhythm Classification

The PTB-XL dataset serves as a multi-class classification problem. Unlike the financial task which requires detecting stochastic trends, this task serves as our benchmark for detecting periodic abnormalities. The classification heads must determine if the foundation model has preserved specific waveform information (such as P-wave onset or QRS width) required to distinguish between diagnostic classes like Myocardial Infarction and Hypertrophy.

Classifier	Metric	Wav2Vec	Astromer	FinCast	ECGFounder	TimesFM	Moirai	Custom	Random
k-NN	Accuracy	0.387	0.495	0.314	0.545	0.523	<u>0.540</u>	0.482	0.326
	Precision	0.346	0.462	0.278	0.510	0.495	<u>0.507</u>	0.451	0.287
	Recall	0.387	0.495	0.314	0.545	0.523	<u>0.540</u>	0.482	0.326
	F1	0.353	0.468	0.291	0.517	0.501	<u>0.509</u>	0.460	0.300
Logistic	Accuracy	0.436	0.552	0.437	<u>0.608</u>	0.590	0.617	0.534	0.437
	Precision	0.354	0.517	0.191	<u>0.580</u>	0.562	0.594	0.495	0.191
	Recall	0.436	0.552	0.437	<u>0.608</u>	0.590	0.617	0.534	0.437
	F1	0.320	0.504	0.266	<u>0.578</u>	0.560	0.587	0.477	0.266
Centroid	Accuracy	0.206	0.338	0.147	0.435	0.361	0.363	<u>0.395</u>	0.105
	Precision	0.357	0.457	0.274	0.527	<u>0.512</u>	0.498	0.483	0.268
	Recall	0.206	0.338	0.147	0.435	0.361	0.363	<u>0.395</u>	0.105
	F1	0.197	0.350	0.097	0.452	0.400	0.390	<u>0.421</u>	0.114
MLP	Accuracy	0.470 (0.001)	0.557 (0.003)	0.436 (0.002)	<u>0.635 (0.004)</u>	0.622 (0.006)	0.657 (0.001)	0.557 (0.003)	0.385 (0.007)
	Precision	0.410 (0.009)	0.524 (0.003)	0.230 (0.032)	<u>0.619 (0.005)</u>	0.601 (0.007)	0.641 (0.002)	0.527 (0.006)	0.272 (0.003)
	Recall	0.470 (0.001)	0.557 (0.003)	0.436 (0.002)	<u>0.635 (0.004)</u>	0.622 (0.006)	0.657 (0.001)	0.557 (0.003)	0.385 (0.007)
	F1	0.384 (0.008)	0.507 (0.003)	0.271 (0.004)	<u>0.622 (0.005)</u>	0.600 (0.005)	0.639 (0.003)	0.510 (0.003)	0.294 (0.001)

Table 10: Classification results for PTB-XL dataset. Best results are **bold**, second best are underlined. The MLP are run with 3 seeds (0, 1, 2) and results are reported as Mean (Standard Deviation).

D.3 Acoustic Classification

The ESC-50 dataset benchmarks the ability for a model to understand frequency domain of our data. Due to the nature of audio recordings, the embeddings must implicitly capture spectral decomposition information,

mapping waveforms to frequency signatures, to distinguish between acoustically similar classes (e.g., ‘Rain’ vs. ‘Wind’) without explicit Fourier transform preprocessing.

Classifier	Metric	Wav2Vec	Astromer	FinCast	ECGFounder	TimesFM	Moirai	Custom	Random
k-NN	Accuracy	0.142	0.072	0.018	<u>0.343</u>	0.282	0.217	0.347	0.020
	Precision	0.173	0.090	0.004	<u>0.332</u>	0.286	0.206	0.378	0.020
	Recall	0.142	0.072	0.018	<u>0.343</u>	0.282	0.217	0.347	0.020
	F1	0.132	0.069	0.004	<u>0.320</u>	0.256	0.200	0.331	0.017
Logistic	Accuracy	0.282	0.158	0.018	<u>0.422</u>	0.370	0.328	0.455	0.033
	Precision	0.291	0.119	0.002	<u>0.439</u>	0.375	0.355	0.457	0.031
	Recall	0.282	0.158	0.018	<u>0.422</u>	0.370	0.328	0.455	0.033
	F1	0.278	0.127	0.003	<u>0.407</u>	0.357	0.325	0.441	0.031
Centroid	Accuracy	0.175	0.075	0.020	0.388	0.200	0.207	<u>0.323</u>	0.033
	Precision	0.153	0.045	0.001	0.348	0.170	0.234	<u>0.316</u>	0.033
	Recall	0.175	0.075	0.020	0.388	0.200	0.207	<u>0.323</u>	0.033
	F1	0.139	0.044	0.002	0.349	0.162	0.190	<u>0.298</u>	0.032
MLP	Accuracy	0.257 (0.006)	0.090 (0.002)	0.023 (0.003)	0.475 (0.004)	0.272 (0.008)	0.254 (0.010)	<u>0.317 (0.008)</u>	0.026 (0.006)
	Precision	0.257 (0.009)	0.075 (0.003)	0.008 (0.010)	0.469 (0.007)	0.259 (0.013)	0.249 (0.006)	<u>0.305 (0.008)</u>	0.026 (0.006)
	Recall	0.257 (0.006)	0.090 (0.002)	0.023 (0.003)	0.475 (0.004)	0.272 (0.008)	0.254 (0.010)	<u>0.317 (0.008)</u>	0.026 (0.006)
	F1	0.244 (0.007)	0.066 (0.001)	0.004 (0.003)	0.456 (0.004)	0.244 (0.009)	0.232 (0.006)	<u>0.289 (0.006)</u>	0.025 (0.006)

Table 11: Classification results for ESC-50 dataset. Best results are **bold**, second best are underlined. The MLP are run with 3 seeds (0, 1, 2) and results are reported as Mean (Standard Deviation).

D.4 Light curve Classification

To test a model’s ability to encode deterministic physical laws, we utilize the StarEmbed dataset. Unlike the previous datasets, light curves are determined by strict astrophysical periodicity. The classification heads must determine if the embedding can capture the periodic nature of these lightcurves despite irregular sampling.

Classifier	Metric	Wav2Vec	Astromer	FinCast	ECGFounder	TimesFM	Moirai	Custom	Random
k-NN	Accuracy	0.649	<u>0.661</u>	0.650	0.648	0.648	0.651	0.888	0.649
	Precision	0.488	<u>0.555</u>	0.483	0.482	0.491	0.493	0.885	0.487
	Recall	0.649	<u>0.661</u>	0.650	0.648	0.648	0.651	0.888	0.649
	F1	0.543	<u>0.581</u>	0.543	0.543	0.543	0.545	0.886	0.543
Logistic	Accuracy	<u>0.676</u>	0.675	<u>0.676</u>	<u>0.676</u>	<u>0.676</u>	<u>0.676</u>	0.894	<u>0.676</u>
	Precision	0.457	<u>0.508</u>	0.457	0.457	0.457	0.457	0.890	0.457
	Recall	<u>0.676</u>	0.675	<u>0.676</u>	<u>0.676</u>	<u>0.676</u>	<u>0.676</u>	0.894	<u>0.676</u>
	F1	0.545	<u>0.552</u>	0.545	0.545	0.545	0.545	0.890	0.545
Centroid	Accuracy	0.089	<u>0.133</u>	0.077	0.048	0.032	0.043	0.495	0.062
	Precision	0.499	<u>0.519</u>	0.469	0.504	0.459	0.446	0.791	0.476
	Recall	0.089	<u>0.133</u>	0.077	0.048	0.032	0.043	0.495	0.062
	F1	0.122	<u>0.159</u>	0.092	0.053	0.027	0.045	0.524	0.077
MLP	Accuracy	0.667 (0.001)	<u>0.679 (0.001)</u>	0.674 (0.002)	0.590 (0.008)	0.670 (0.007)	0.676 (0.000)	0.903 (0.000)	0.662 (0.001)
	Precision	0.494 (0.017)	<u>0.563 (0.014)</u>	0.469 (0.017)	0.481 (0.003)	0.473 (0.013)	0.457 (0.000)	0.902 (0.001)	0.483 (0.003)
	Recall	0.667 (0.001)	<u>0.679 (0.001)</u>	0.674 (0.002)	0.590 (0.008)	0.670 (0.007)	0.676 (0.000)	0.903 (0.000)	0.662 (0.001)
	F1	0.546 (0.000)	<u>0.577 (0.005)</u>	0.545 (0.001)	0.526 (0.004)	0.545 (0.000)	0.545 (0.000)	0.901 (0.001)	0.544 (0.000)

Table 12: Classification results for StarEmbed dataset. Best results are **bold**, second best are underlined. The MLP are run with 3 seeds (0, 1, 2) and results are reported as Mean (Standard Deviation).

D.5 Full Geometric Analysis

Domain	Model	Embed Size	Intrinsic Dim	Anisotropy
Finance (S&P 500)	Random	512	159.9	0.750
	Astromer	256	9.3	1.000
	Custom	18	5.3	0.951
	ECGFounder	1024	12.5	0.428
	FinCast	1280	4.0	0.999
	Moirai	384	11.2	0.748
	TimesFM	1280	14.0	0.986
	Wav2Vec	768	10.6	0.859
Astronomy (Star)	Random	512	146.6	0.751
	Astromer	513	4.9	0.996
	Custom	54	5.6	0.775
	ECGFounder	2048	14.5	0.628
	FinCast	2561	4.3	0.997
	Moirai	768	10.0	0.838
	TimesFM	2560	9.8	0.997
	Wav2Vec	1536	11.8	0.825
Medical (PTB-XL)	Random	512	136.9	0.750
	Astromer	256	5.8	0.999
	Custom	18	3.4	0.809
	ECGFounder	1024	19.6	0.757
	FinCast	1280	14.5	1.000
	Moirai	384	13.7	0.970
	TimesFM	1280	12.3	0.999
	Wav2Vec	768	10.7	0.751
Audio (ESC-50)	Random	512	122.5	0.750
	Astromer	257	1.5	0.986
	Custom	36	3.5	0.743
	ECGFounder	1025	4.9	0.996
	FinCast	1280	3.0	0.984
	Moirai	385	2.0	1.000
	TimesFM	1281	3.0	0.904
	Wav2Vec	769	2.5	0.998

Table 13: **Full Geometric Analysis Results.** Detailed topological metrics for all evaluated models across four domains.